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RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
IV Semester B.E. Sep -2024 Examinations
DEPARTMENT ELECTRONICS AND COMMUNICATION ENGINEERING
Model Question Paper
TRANSMISSION LINES AND ELECTROMAGNETIC FIELDS
(2022 SCHEME)

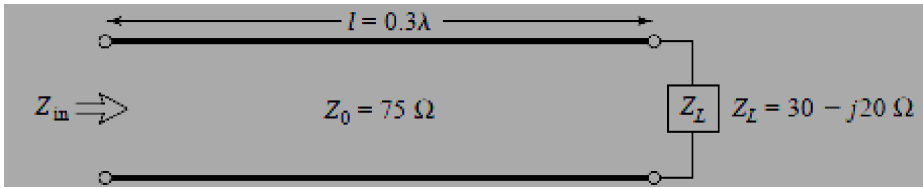
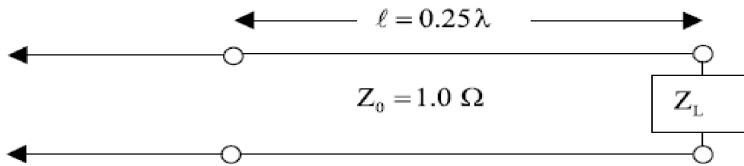
Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

		PART-A (Objective type for one or two marks) (True & false and match the following questions are not permitted)	
1	1.1	An open wire transmission line terminated in its primary constant at 1KHz $R=4\Omega/\text{Km}$, $L=3.5\text{mH}/\text{Km}$, $C=0.009\mu\text{F}/\text{Km}$, $G=0.29\mu/\Omega\text{Km}$ then characteristic impedance of line is	02
	1.2	If the primary constants of a telephone line are $R=10\Omega/\text{Km}$, $C=0.04\mu\text{F}/\text{Km}$ operating at 1KHz, then the attenuation constant is _____. If the telephone line is loaded with coils at regular intervals which lead to an inductance of $111.11\text{mH}/\text{Km}$ and the resistive losses are $23.3\Omega/\text{Km}$, then the attenuation constant of the line is	02
	1.3	If $D = z\rho\cos^2\phi\hat{x}\text{ c/m}^2$. Find Charge density at $(1, \pi/2, 3)$. Also find charge enclosed at $r=1\text{m}$, $-2 < z < 2\text{m}$.	02
	1.4	Let $V = xy^2z$. find the energy expended in transferring a $5\mu\text{C}$ point charge from $(1, -1, 2)$ to $(2, 1, -3)$.	02
	1.5	The xy-plane serves as the interface between two different media. Medium 1 ($z < 0$) is filled with a material μ_{r1} , and medium 2 ($z > 0$) is filled with a material μ_{r2} . If the interface carries current $K\text{ y mA/m}$, then Magnetic field tangential component and normal components are and	02
	1.6	The magnetic vector potential $A = 15e^{-\rho}\sin(\Phi)z$, find H at $(4, \pi/4, -10)$ and flux through the surface $\rho = 4$, $0 < \Phi < \pi/2$, $0 < z < 10$.	02
	1.7	For a circular wire radius is 1.5mm with current $i = 5.5\sin\sin(4 \times 10^8 t)\mu\text{A}$ with conductivity $\sigma=35\text{ mho/m}$ and $\epsilon_r=10$. Find displacement current density.	02
	1.8	Write Maxwell's Equations in Point form	02
	1.9	In a lossless dielectric $\eta=60\pi\Omega$, $\mu_r=1$ $H = -0.1\cos(\omega t - z)\hat{x}\text{ A/m}$ Find ϵ , ω	02
	1.10	Define skin depth and transmission coefficient	02
Unit 1			
2	a	An open wire transmission line having primary constants $R\text{ ohm/m}$, $L\text{ H/m}$, $G\text{ mho/m}$ and $C\text{ F/m}$ operating at $f\text{ Hz}$ then i. Derive an expression for transmission line using distributed parameter model i. Bring out the similarities between Transmission Line Equation and Helmholtz's wave equation ii. If $V=V_s$ and $I=I_s$ at generator end Derive an expression to determine Voltage and current at any point l along the line	08
	b	The open circuit and short circuit impedance at 10000 rad/sec of a 500Km length transmission line are respectively $Z_{oc}=2000\angle -80^\circ\Omega$ and $Z_{sc}=20\angle 20^\circ\Omega$. Determine the	08

		characteristic impedance, Propagation constant, Phase constant and the attenuation constant of the line.	
Unit 2			
3	a	A lossless transmission line of electrical length $l = 0.3\lambda$ is terminated with a complex load impedance as shown in the Fig. 1.  Fig. 1. Find - The reflection coefficient at the load, - The SWR on the line, - The input impedance to the line.	08
	b	Find D at the origin if the following charge distributions are present in free space: point charge, 12 nC at P (2, 0, 6); uniform line charge density, 3nC/m at x = -2, y = 3; uniform surface charge density, 0.2 nC/m ² at x = 2.	08
OR			
4	a	The transmission line shown in Fig. 3 has load impedance $0.2 - j0.2 \Omega$. (Solve using Smith Chart Only).  Fig. 3. - Find Impedance Z at $l = 0.25\lambda$ - Find VSWR on the line - Find reflection coefficient of the line How far from the load is first voltage minimum	08
	b	Two point charges of $-1\mu\text{C}$ is located at P1(0, 0, 0.5) and P2(0, 0, -0.5), and a third charge of $2\mu\text{C}$ is at the origin, find Electric field E(0, 2, 1) in spherical co-ordinate systems.	08
Unit 3			
5	a	Conducting spherical shells with radii $a = 10\text{cm}$ and $b = 30\text{cm}$ are maintained at a potential difference of 100V such that $V(r=b) = 0$ and $V(r=a) = 100\text{V}$. Determine V and E in the region between the shells. Determine the maximum E. If $\epsilon_r = 2.5$ in the region, determine the surface charge density on the shells and the capacitance of the capacitor.	08
	b	Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z \geq 0$, $\epsilon_{r1} = 4$ and for $z \leq 0$, $\epsilon_{r2} = 3$. A uniform electric field $E_1 = 5ax - 2ay + 3az \text{ kV/m}$ exists for $z \geq 0$. Find E2 for $z \leq 0$, the angles E1 and E2 make with interface, the	08

		energy densities in both dielectrics, and the energy within a cube of side 2 m centered at (3, 4, and 5).	
		OR	
6	a	Let the region $Z < 0$ be composed of uniform dielectric material for which $\epsilon_r = 3.2$ while the region $Z > 0$ is characterized by $\epsilon_r = 2$. Let $D_1 = -30a_x + 50a_y + 70a_z \text{ nC/m}^2$. Find D_2, E_2 - Angles between D_2 with interface and D_1 with interface - Energy density in region 1&2.	08
	b	A charge distribution with spherical symmetry has density $\rho_v = \begin{cases} \frac{r}{R} \rho_0 & 0 \leq r \leq R \\ 0 & \text{elsewhere} \end{cases}$ - Find V for $r > R$ - Find V for $r \leq R$ - Find the total charge	08
Unit 4			
7	a	Let $\mu_{R1} = 2$ in region 1, defined by $2x+3y-4z > 1$, while $\mu_{R2} = 5$ in region 2 where $2x+3y-4z < 1$. In region 1, $H_1 = 50a_x - 30a_y + 20a_z \text{ A/m}$. Find: H_{N1} H_{T1} H_{T2} H_{N2} θ_1, the angle between H_1 and a_{N21}	10
	b	Assume that $A = 50\rho^2 a_z \text{ Wb/m}$ in a certain region of free space. - Find H and B - Find J - Use J to find the total current crossing the surface $0 \leq \rho \leq 1, 0 \leq \phi < 2\pi, z = 0$	06
		OR	
8	a	Given the field $H = 20\rho^2 \hat{\phi} \text{ A/m}$: - Determine the current density J, - Determine total current that crosses surface $\rho = 1, 0 < \phi < 2\pi, z = 0$:	08
	b	An infinitely long transmission line consisting of two concentric cylinders having their axes along the z-axis. The inner conductor has radius a and carries current I while the outer conductor has inner radius b and thickness t and carries return current $-I$. Determine H everywhere assuming that current is uniformly distributed in both conductors	08

Unit 5

9	a	Derive equation of continuity. - Using the expression derived in i. show that Ampere circuital law cannot be applicable for time varying currents. - Modify Ampere circuital law such that it is applicable for time varying currents.	08
	b	A uniform plane wave in air is normally incident on an infinite lossless dielectric material having $\epsilon = 3\epsilon_0$ and $\mu = \mu_0$. If the incident wave is $\mathbf{E}_i = 10 \cos(\omega t - z) \hat{\mathbf{y}} \text{ V/m}$ find: - Wavelength λ and angular frequency ω of the wave in air. - The transmitted wave in the dielectric medium. - The incident Magnetic field. - Reflection coefficient and transmitted coefficient. - The total electric field in air medium. The total electric field in dielectric medium.	08

		OR	
10	a	State the Poynting theorem and derive the expression to prove the Poynting theorem	08
	b	A uniform plane wave propagating in a medium has $E = 2e^{-\alpha z} \sin(10^8 t - \beta z) \hat{y} \text{ V/m}$ find: - Attenuation constant α - Phase constant β - Wave polarization - Magnetic field $\mathbf{H}$	08

Signature of Scrutinizer:

Signature of Chairman

Name:

Name: